

Question

The wastewater produced during copper smelting contains a high concentration of arsenic, mainly in the form of AsO_2^- . The following treatments were all carried out at room temperature.

First, add a 26% Na_2S solution, adjust the pH to 0.8 with sulfuric acid, and after reacting for twenty minutes, a precipitate is obtained, which is recorded as reaction 1.

Subsequently, the precipitate is added to a slightly excessive amount of $NaOH$ solution at a molar ratio of 1 : 7 to ensure complete dissolution of the precipitate, which is recorded as reaction 2.

After dissolution, a small amount of hydroquinone is added as a catalyst and sodium dodecyl sulfonate as a surfactant, and air is passed through for ten hours for oxidation and desulfurization, which is recorded as reaction 3.

After oxidation, the pH is adjusted to 0 with sulfuric acid, and SO_2 is then passed through to generate a precipitate, which is recorded as reaction 4.

Finally, the precipitate is filtered to obtain a solid oxide M with high purity. Detection reveals that the final remaining filtrate still contains a certain amount of Re element that can be recovered.

The above filtrate is extracted with an organic phase composed of 20% trioctylamine, 40% sec-octyl alcohol, and 40% kerosene. After liquid separation, the organic phase is taken and extracted with ammonia water. After liquid separation again, the aqueous phase is frozen, crystallized, and separated to obtain a pure compound A without water of crystallization.

1.3874g of compound A is taken and gradually heated to $800^\circ C$ in a H_2 atmosphere, and the final solid product B is 0.9631g Re , and 0.3727g of water vapor is separated in the tail gas. If heated in an air atmosphere, 100% weight loss is observed at $400^\circ C$.

Regarding the above process, the following descriptions are given. Let the product of the serial numbers of all incorrect descriptions be x , and the sum of the serial numbers of all correct descriptions be y . Find $\frac{x}{(y+10)^2}$ and select the option with an absolute error < 0.001 . If there is no option that meets the condition, select A .

1. It is not possible to select four coefficients from the equation for reaction 1 that form an arithmetic sequence.
2. The sum of the coefficients on the left side of the equation for reaction 2 divided by the sum of the coefficients on the right side is equal to 1.4.
3. The average coefficient of the equation for reaction 3 is equal to 3.8.
4. The product of all the coefficients on both sides of the equation for reaction 4 is equal to 144.
5. The catalytic principle of hydroquinone is the generation of a p-benzoquinone intermediate.

6. One molecule of compound A contains 10 atoms.
7. One molecule of compound A contains 9 atoms.
8. When compound A is treated at 800°C in a H_2 atmosphere, one molecule of compound A generates 4 molecules of gas.
9. When treated at 400°C in an air atmosphere, a reaction with oxygen occurs.
10. Organic phase extraction utilizes sec-octyl alcohol and kerosene to suppress the formation of a third phase.
11. Back extraction cannot use a high pH solution.

A. All other options are incorrect

B. 1.429

C. 0.802

D. 1.565

E. 0.913

F. 1.905

G. 0.634

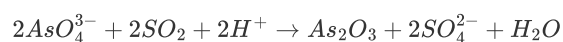
H. 2.665

Answer

Correct Answer: **E**

Detailed Explanation

The equations for reactions 1 to 4 are as follows:



The coefficients 1, 2, 3, 4 in the equation of reaction 1 constitute an arithmetic sequence with a common difference of 1, so statement 1 is incorrect.

CHECKPOINT

0.5 PTS

Equation for reaction 1 $2AsO_2^- + 3S^{2-} + 8H^+ \rightarrow As_2S_3 + 4H_2O$

The sum of the coefficients on the left side of the equation for reaction 2, 7, divided by the sum of the coefficients on the right side, 5, equals 1.4, so statement 2 is correct.

CHECKPOINT

0.5 PTS

Equation for reaction 2 $As_2S_3 + 6OH^- \rightarrow AsO_3^{3-} + AsS_3^{3-} + 3H_2O$

The average coefficient of the equation for reaction 3 is $\frac{2+2+5+4+6}{5} = 3.8$, so statement 3 is correct.

CHECKPOINT

0.5 PTS

Equation for reaction 3 $2AsO_3^{3-} + 2AsS_3^{3-} + 5O_2 \rightarrow 4AsO_4^{3-} + 6S$

The product of all the coefficients on both sides of the equation for reaction 4 is equal to 16, so statement 4 is incorrect.

CHECKPOINT

0.5 PTS

Equation for reaction 4 $2AsO_4^{3-} + 2SO_2 + 2H^+ \rightarrow As_2O_3 + 2SO_4^{2-} + H_2O$

The first step of hydroquinone catalysis is oxidation to benzoquinone by oxygen, so statement 5 is correct.

CHECKPOINT

0.5 PTS

In the first step of catalysis, hydroquinone is oxidized to benzoquinone

The molar mass of compound A is $M_A = \frac{m_A Ar(Re)}{m_B} = \frac{1.3874 \times 186.2}{0.9631} = 268.2 \text{ g/mol}$

The molar ratio of Re and O in compound A is $n(Re) : n(O) = \frac{m_B}{Ar(Re)} : \frac{m_{H_2O}}{Mr(H_2O)} = \frac{0.9631}{186.2} : \frac{0.3727}{18.02} = 1 : 3.999 \approx 1 : 4$

The remaining element $Mr = 268.2 - 186.2 - 16.00 \times 4 = 18.0$, which is inferred to be NH_4^+

Therefore, the chemical formula of compound A is NH_4ReO_4 , which contains 10 atoms in one molecule, so statement 6 is correct and statement 7 is incorrect.

CHECKPOINT

1 PTS

Compound A is NH_4ReO_4

The equation for treatment at 800°C under H_2 atmosphere is $2NH_4ReO_4 + 7H_2 \rightarrow 2NH_3 + 2Re + 8H_2O$, so one molecule of A generates 5 molecules of gas, so statement 8 is incorrect.

CHECKPOINT

0.5 PTS

Hydrogen atmosphere equation $2NH_4ReO_4 + 7H_2 \rightarrow 2NH_3 + 2Re + 8H_2O$

The equation for treatment at 400°C under air atmosphere is $2NH_4ReO_4 \rightarrow 2NH_3 + Re_2O_7 + H_2O$, which is the self-decomposition of compound *A*, without the participation of oxygen, so statement 9 is incorrect.

CHECKPOINT

0.5 PTS

Air atmosphere equation $2NH_4ReO_4 \rightarrow 2NH_3 + Re_2O_7 + H_2O$

The equation for organic phase extraction is $R_3N(\text{organic phase}) + ReO_4^-(\text{aqueous phase}) + H^+(\text{aqueous phase}) \rightarrow R_3NH \cdot ReO_4(\text{organic phase})$, in the organic phase, trioctylamine is used to affinity the required product, and sec-octyl alcohol and kerosene are used to enhance the uniformity of the organic phase and avoid the precipitation of the formed ion pairs into a third phase, so statement 10 is correct.

CHECKPOINT

0.5 PTS

sec-octyl alcohol and kerosene avoid the precipitation of the formed ion pairs into a third phase

The equation for ammonia back-extraction is $R_3NH \cdot ReO_4(\text{organic phase}) + NH_3(\text{aqueous phase}) \rightarrow R_3N(\text{organic phase}) + NH_4^+(\text{aqueous phase}) + ReO_4^-(\text{aqueous phase})$, if a high *pH* solution is used, other cations may be introduced or the extraction reaction may be reversed, reducing the recovery rate, so statement 11 is correct.

CHECKPOINT

0.5 PTS

During back-extraction, a high *pH* solution may introduce other cations or reverse the extraction reaction

All correct statements are 2, 3, 5, 6, 10, 11, calculate $y = 2 + 3 + 5 + 6 + 10 + 11 = 37$; all incorrect statements are 1, 4, 7, 8, 9, calculate $x = 1 \times 4 \times 7 \times 8 \times 9 = 2016$. Then $\frac{x}{(y+10)^2} = \frac{2016}{(37+10)^2} = 0.91263 \approx 0.913$, so option *E* is correct.

CHECKPOINT

0.5 PTS

$$x = 2016, y = 37, \frac{x}{(y+10)^2} = 0.913$$